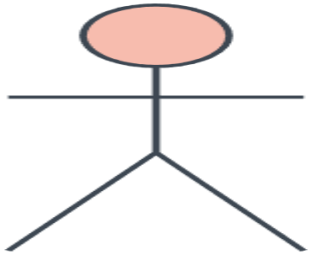
Sequence Diagram

Sequence Diagram

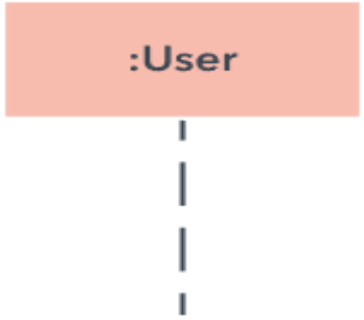
- A sequence diagram is an interaction diagram.
- It represents the flow of message in system.
- Precise description of activities.
- The **sequence diagram consists of**
  - **Objects** represented in the usual way (as named rectangles with the name underlined)
  - **Messages** represented as solid-line arrows
  - **Time** represented as a vertical progression.
  - Fragments

sequence symbols



Actor symbol

Shows entities that interact with or are external to the system.



Lifeline symbol

Represents the passage of time as it extends downward. This dashed vertical line shows the sequential events that occur to an object during the charted process. Lifelines may begin with a labeled rectangle shape or an actor symbol.



Option loop symbol

Used to model if/then scenarios, i.e., a circumstance that will only occur under certain conditions.



Alternative symbol

Symbolizes a choice (that is usually mutually exclusive) between two or more message sequences. To represent alternatives, use the labeled rectangle shape with a dashed line inside.

Message types

Symbol	Name	Description
	Synchronous message symbol	Represented by a solid line with a solid arrowhead. This symbol is used when a sender must wait for a response to a message before it continues. The diagram should show both the call and the reply.
	Asynchronous message symbol	Represented by a solid line with a lined arrowhead. Asynchronous messages don't require a response before the sender continues. Only the call should be included in the diagram.



Asynchronous create message symbol

Represented by a dashed line with a lined arrowhead. This message creates a new object.



Reply message symbol

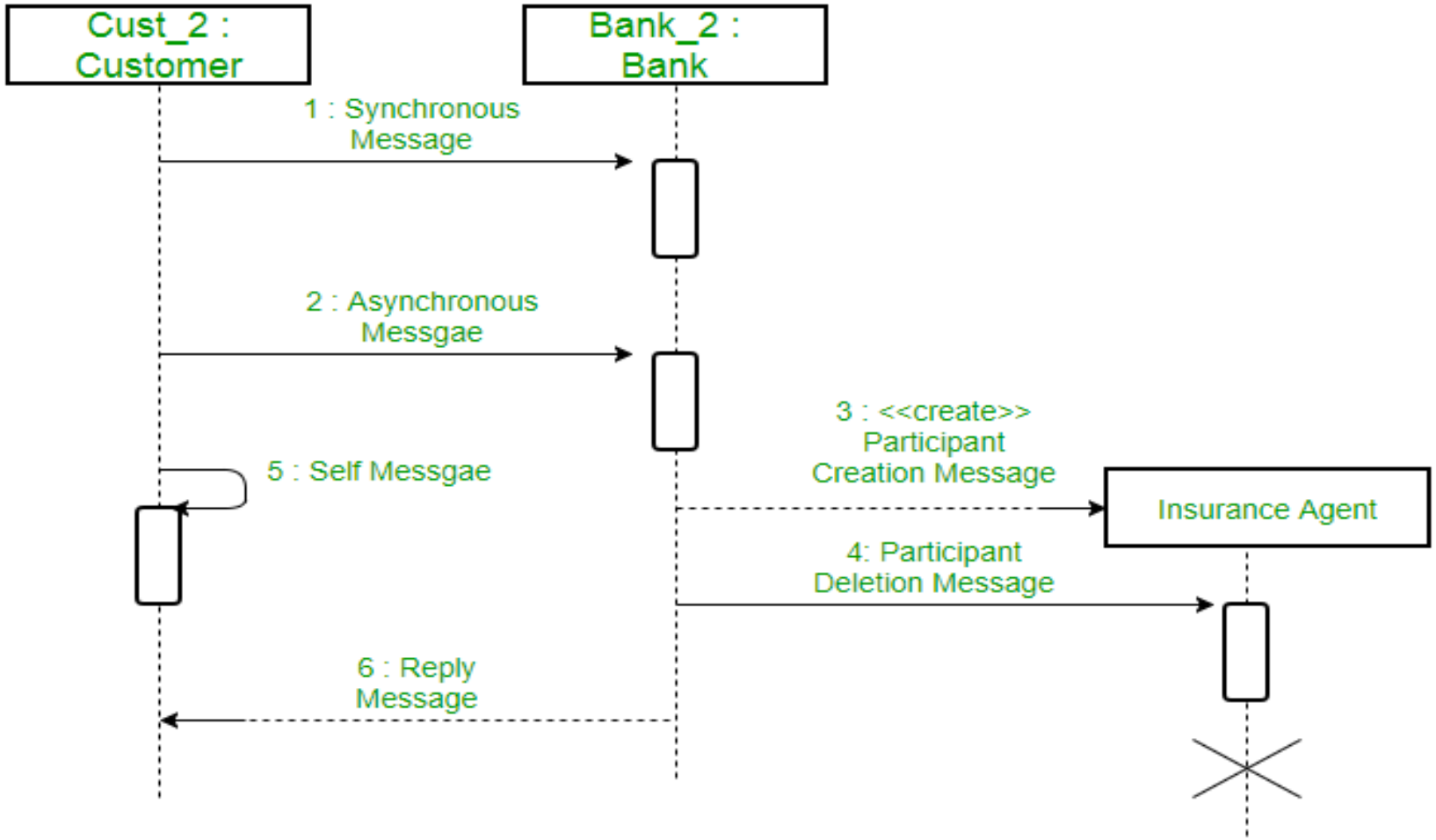
Represented by a dashed line with a lined arrowhead, these messages are replies to calls.



Delete message symbol

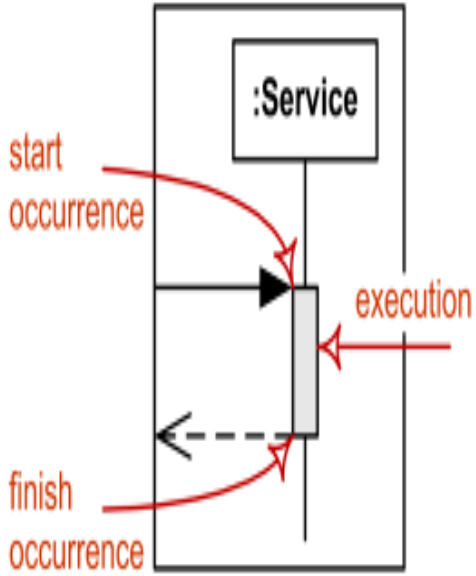
Represented by a solid line with a solid arrowhead, followed by an X. This message destroys an object.

Messages


Cust_2 : Customer
Bank_2 : Bank
1 : Synchronous Message
2 : Asynchronous Messgae
3 : <<create>> Participant Creation Message
Insurance Agent
4: Participant Deletion Message
5 : Self Messgae
6 : Reply Message

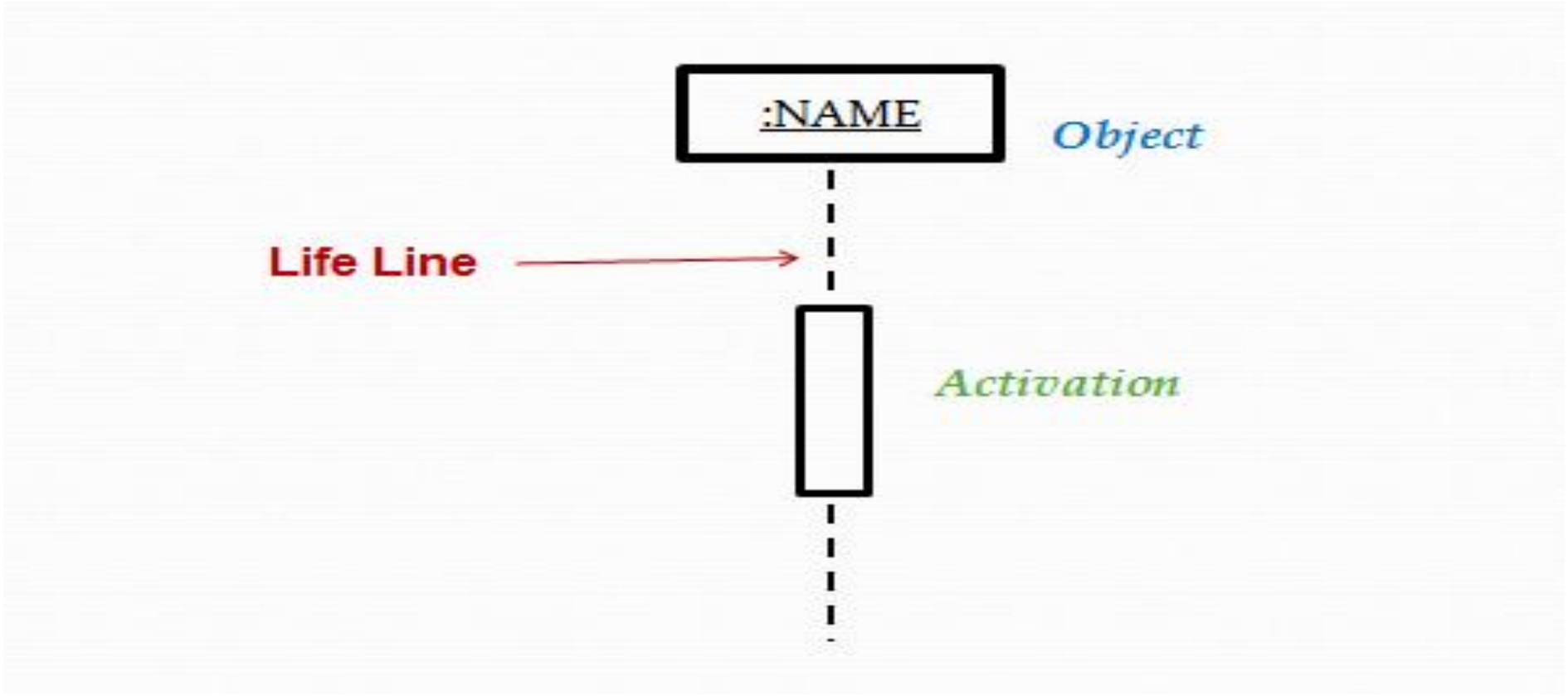
4-Time

- **Execution** (full name - **execution specification**, informally called **activation**) is **interaction fragment** which represents a period in the participant's lifetime when it is
- executing a unit of behavior or action within the **lifeline**,
- sending a signal to another participant,
- waiting for a reply message from another participant.
- The **duration** of an execution is represented by two
- **execution occurrences** - the **start** occurrence and the
- **finish** occurrence.
- **Execution** is represented as a thin grey or white rectangle
- on the lifeline.



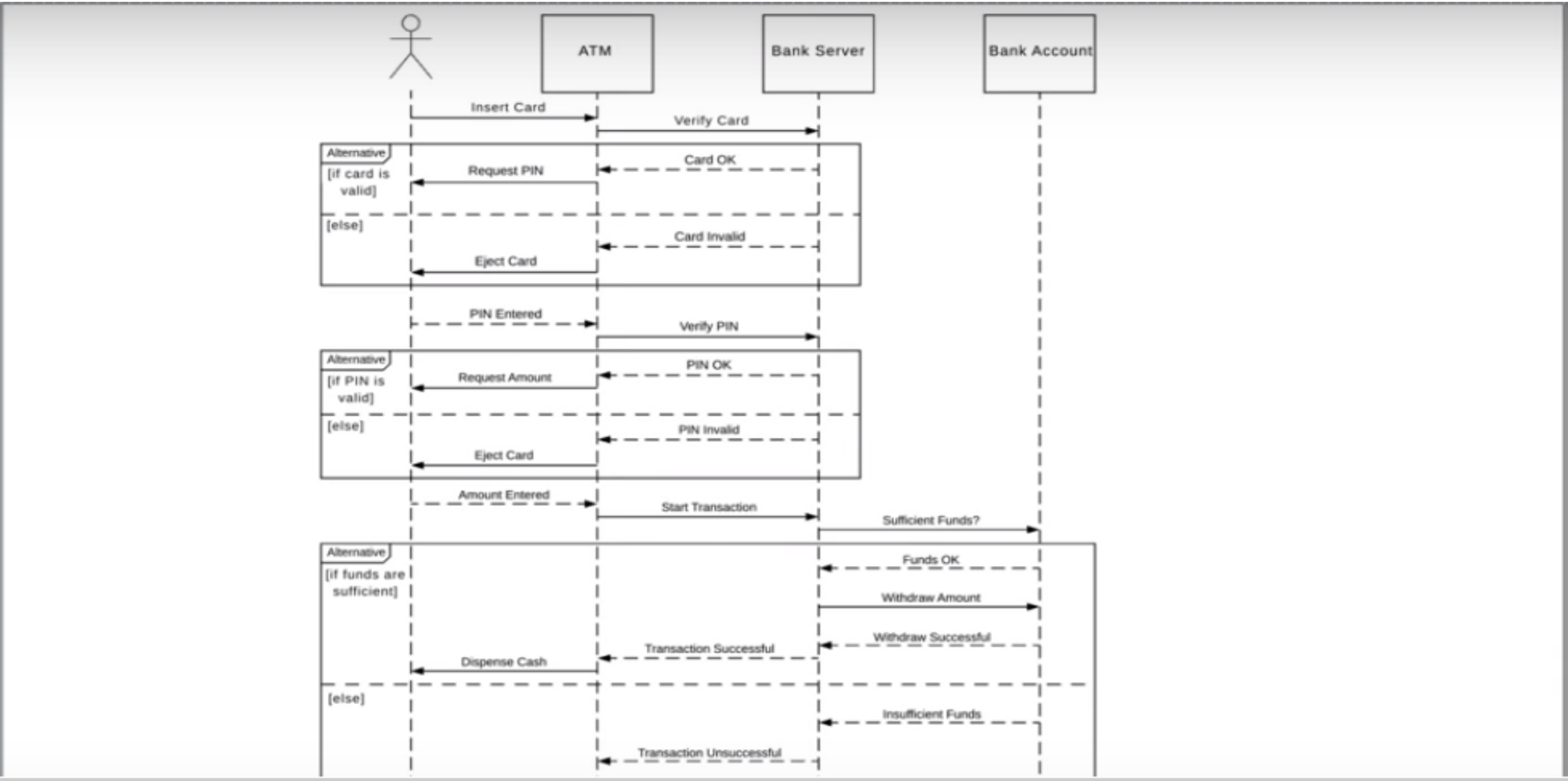
Duration of an execution is represented by two execution occurrences - start and finish.

2-Sequence Diagram



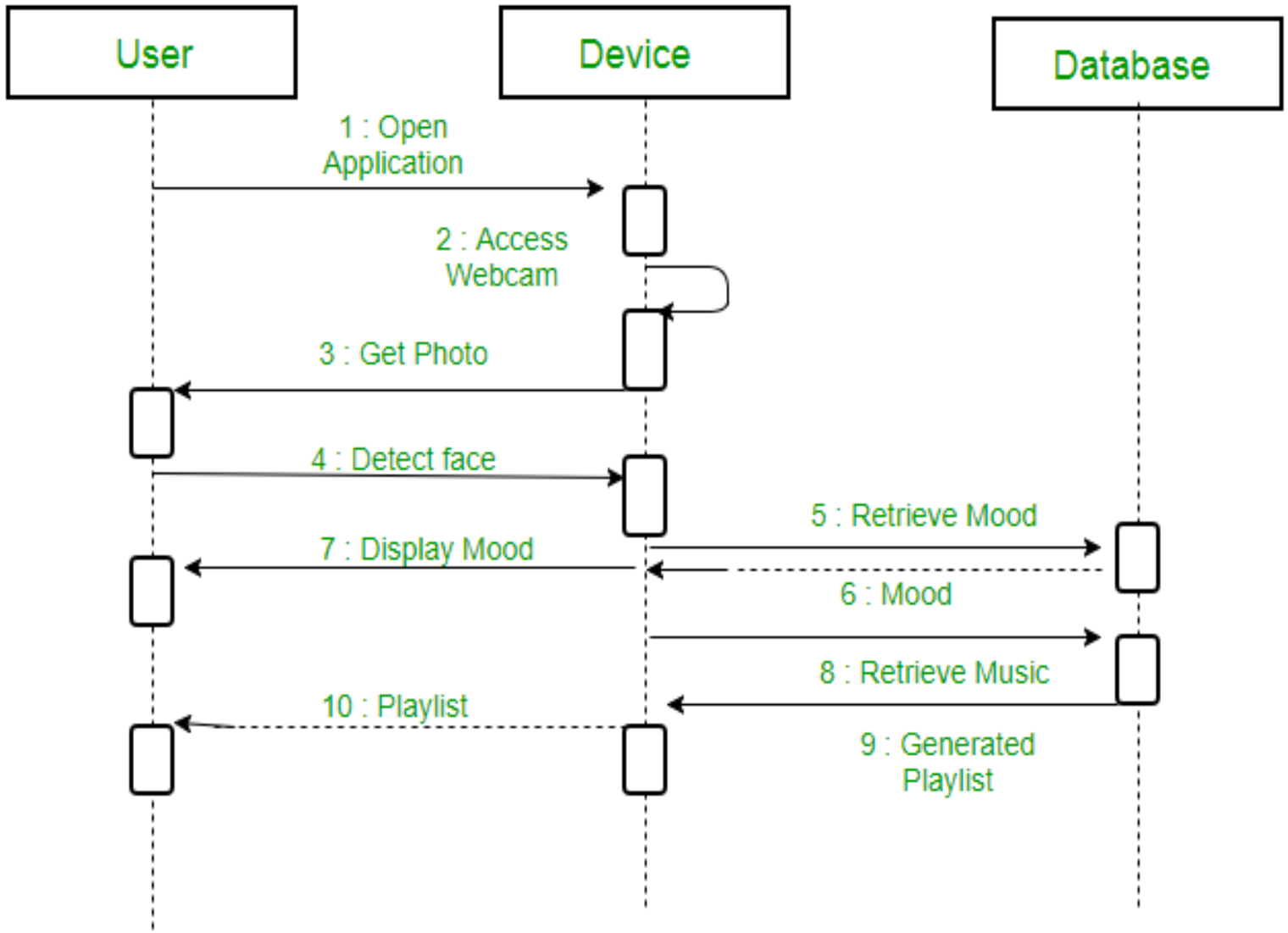
Sequence for ATM

Sequence for ATM

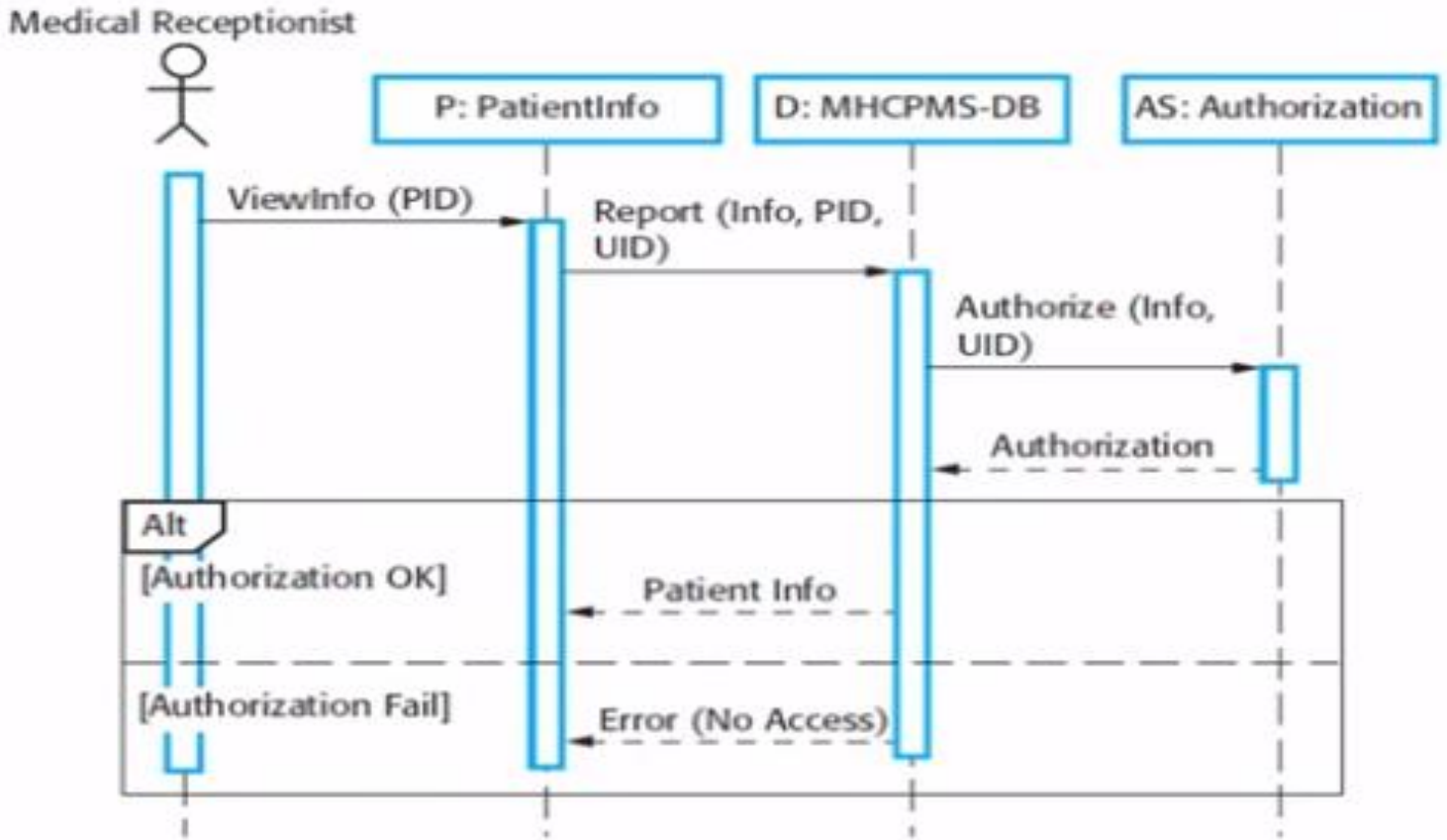


Example: A sequence diagram for an emotion based music player

- Firstly the application is opened by the user.
- The device then gets access to the web cam.
- The webcam captures the image of the user.
- The device uses algorithms to detect the face and predict the mood.
- It then requests database for dictionary of possible moods.
- The mood is retrieved from the database.
- The mood is displayed to the user.
- The music is requested from the database.
- The playlist is generated and finally shown to the user.


User
Device
Database
1 : Open Application
2 : Access Webcam
3 : Get Photo
4 : Detect face
5 : Retrieve Mood
7 : Display Mood
6 : Mood
8 : Retrieve Music
10 : Playlist
9 : Generated Playlist

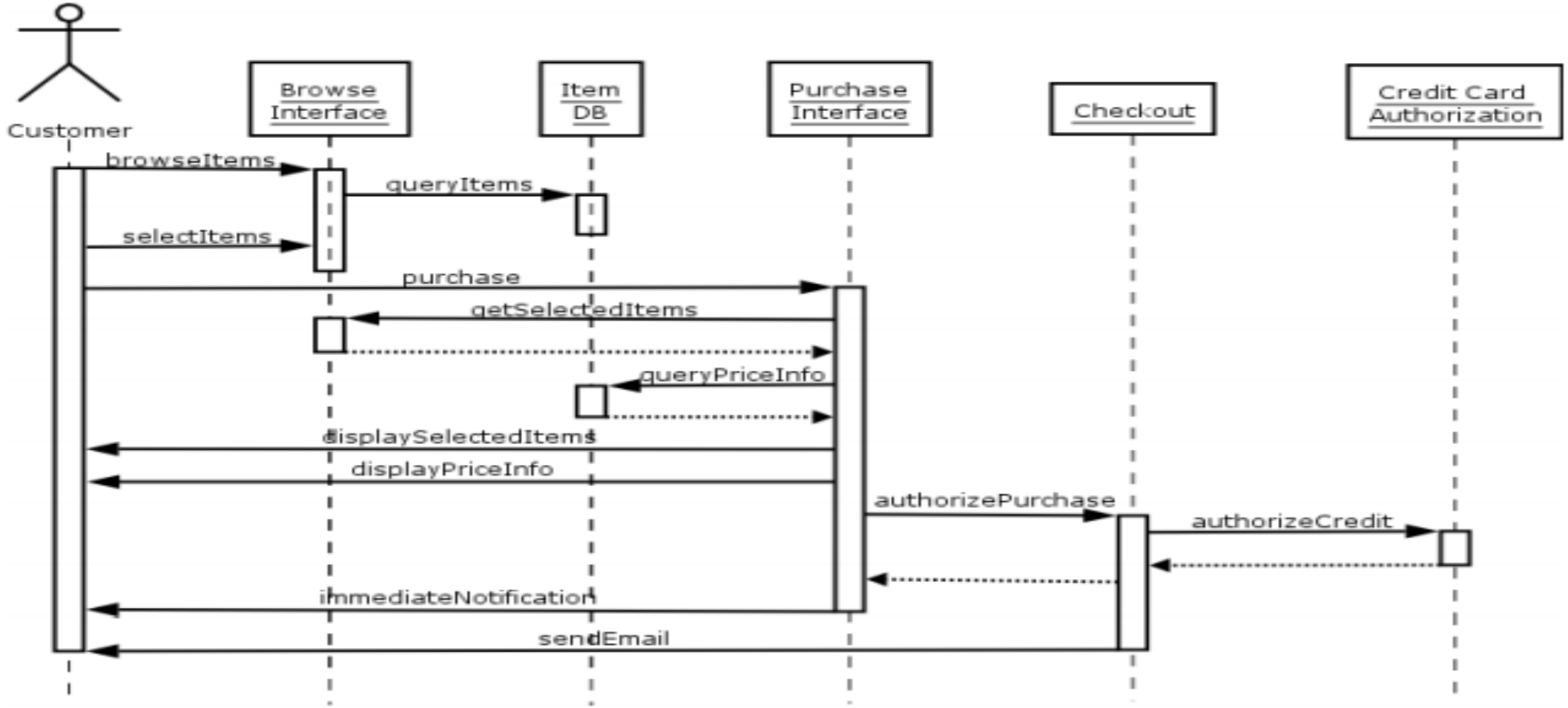
Sequence diagram for View patient information



Example: Online Shopping

- In order to develop our sequence diagram we will need the use case text and the specification class diagram: Use Case Text: Buy a Product Online
  - 1. Customer browses through catalog and selects items to buy.
  - 2. Customer goes to checkout.
  - 3. Customer fills out shipping information.
  - 4. System presents full pricing information, including shipping information.
  - 5. Customer fills in credit card information.
  - 6. System authorizes purchase.
  - 7. System confirms sale immediately.
  - 8. System sends confirming email to customer. The specification class diagram as developed previously:

Sequence Diagram for online shopping



What is State Diagram

- It presents the states an object can be in along with the transitions between the states and shows the starting point and endpoint of a sequence of state changes.

- Examples
    1. Switching light, its state changes from Off to On.
    2. Clicking a remote control, TV changes its state from showing you one channel to showing you another.
    3. After an appropriate amount of time, a washing machine changes its state from Washing to Rinsing.
    4. project: submitted, evaluated, accepted, rejected, monitored, audited, finished, delayed

2-Difference Between State Diagram and Other Diagrams

- The diagrams you've already studied model a **group of classes, objects, or use cases**.

- A **state diagram** shows the states of a **single object.**

State example

- **Example**

Considering the example shown in the above figure, the transition from Waiting state to Riding state takes place when the person gets a taxi. Likewise, the final state is reached, when he reaches the destination. These two occurrences can be termed as events Get_Taxi and Reach_Destination. The following figure shows the events in a state machine.



State machine Diagram example

Car Agency problem

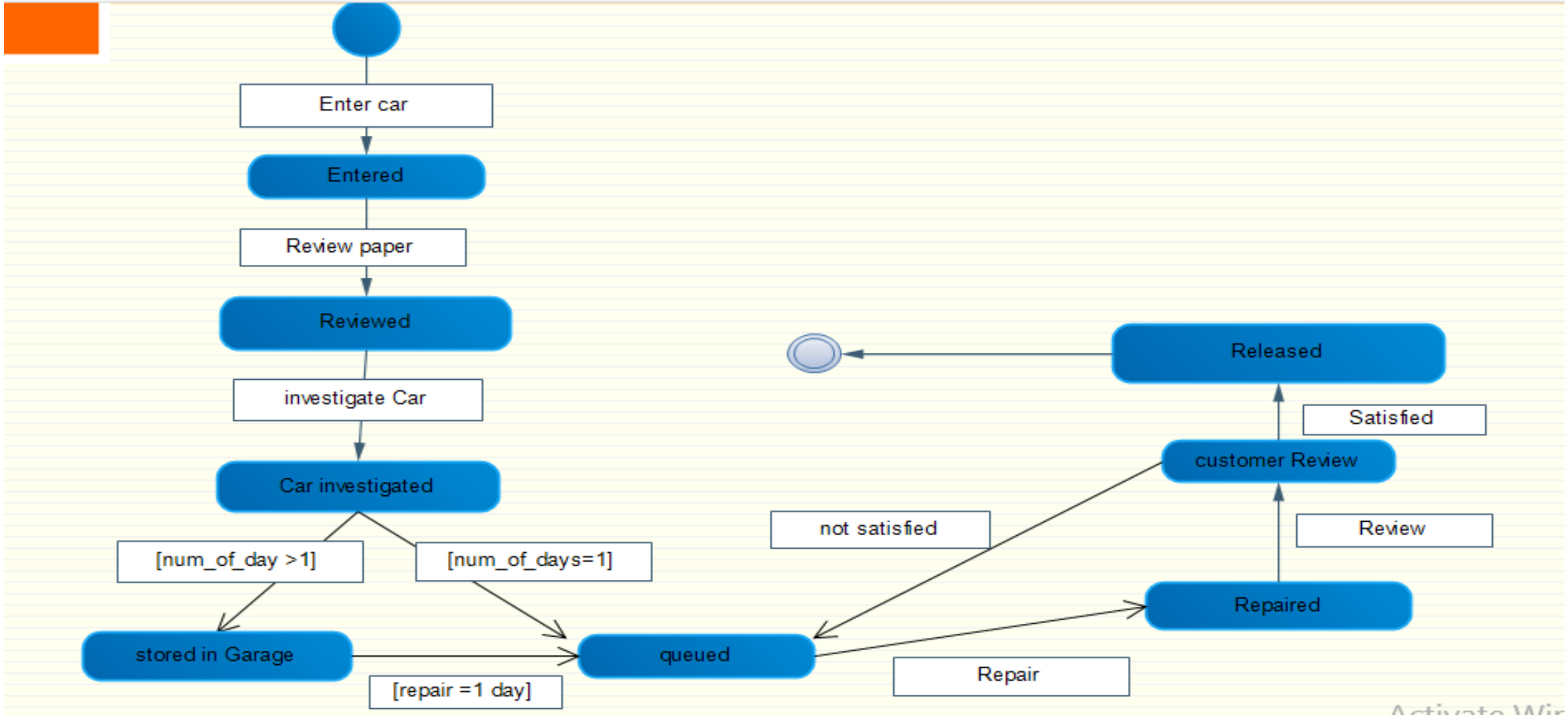
In car Agency : the customer enters the car then car’s papers reviewed after that the car is investigated if the repair take one day

So it will be queued, if it is more than one day it will be stored in Garage.

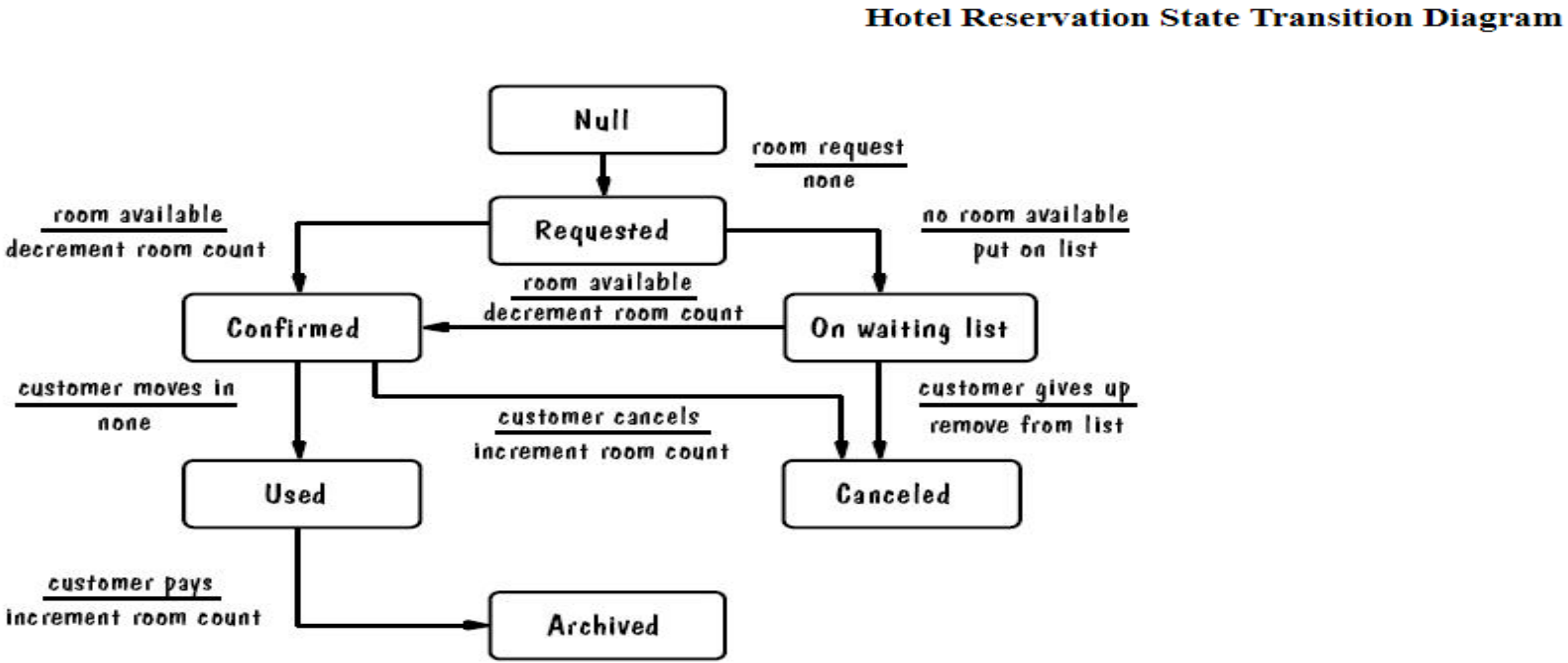
It will be moved from Garage to queue when repair day is today .

It will be moved from the queue to customer check after repaired ,if customer not satisfied it will be queued if satisfied it will be Released.

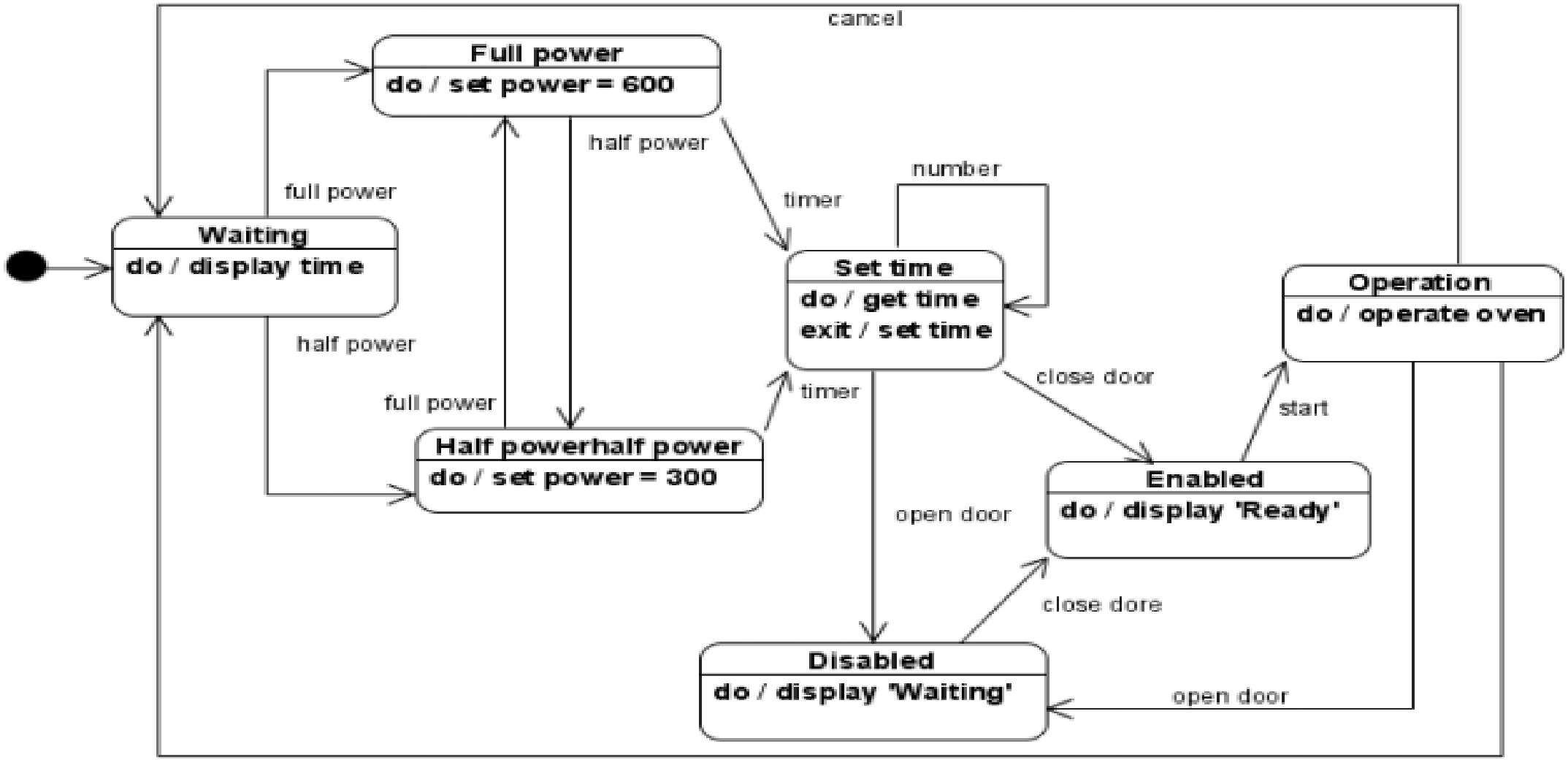
State machine Diagram for car agency



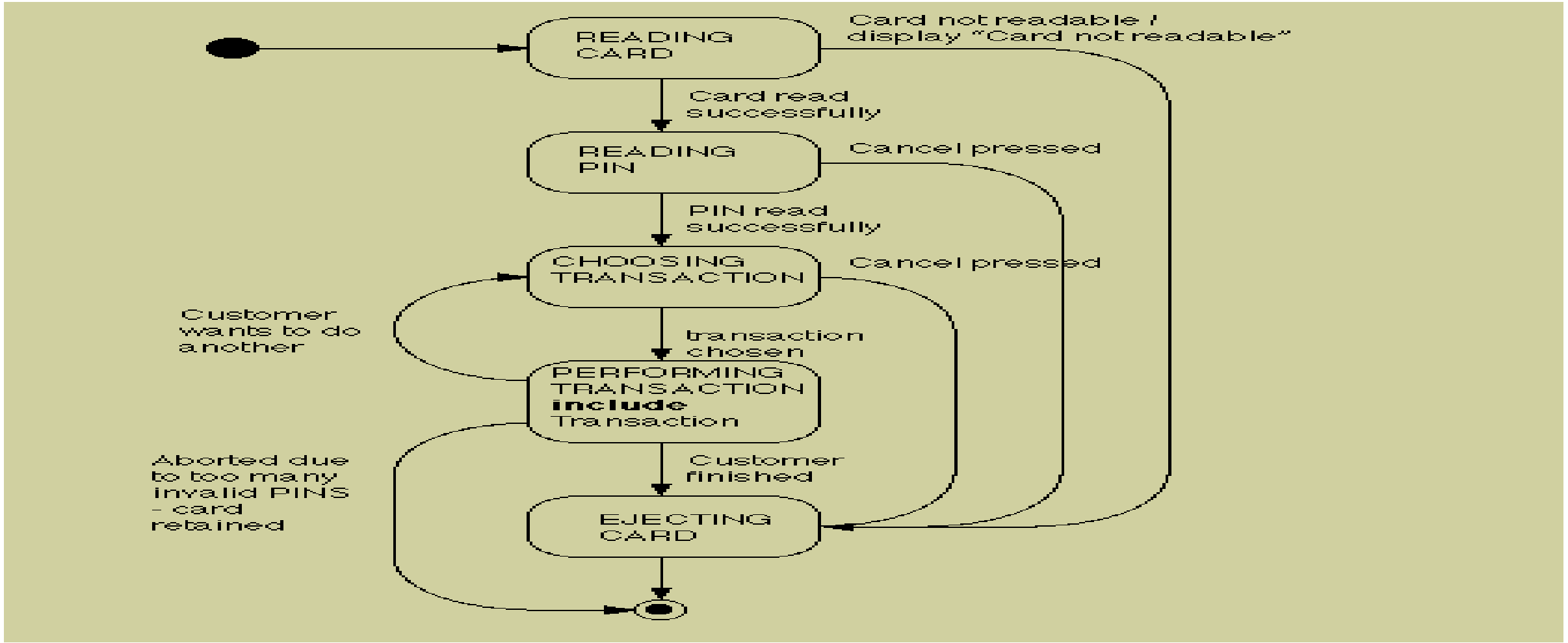
Hotel state machine diagram



State Diagram for microwave oven



State Diagram for ATM



Thank you